

Company2022.com

EmailTrust POC

Advanced POC Results with Client Decisions



Pipl Trust **Proof of Concept**

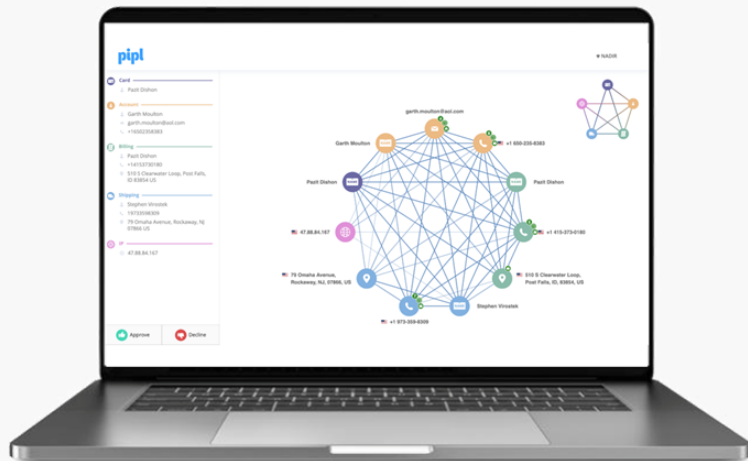
Review of POC Results

pipl.com



Pipl Trust Product Lineup

Pipl provides comprehensive identity intelligence that supports automated decisioning to establish trust, defend against fraud, maximize revenue, and improve customer experience.



Pipl Trust Insights

Innovative manual review tool used by fraud analysts to quickly and accurately determine trusted transactions.

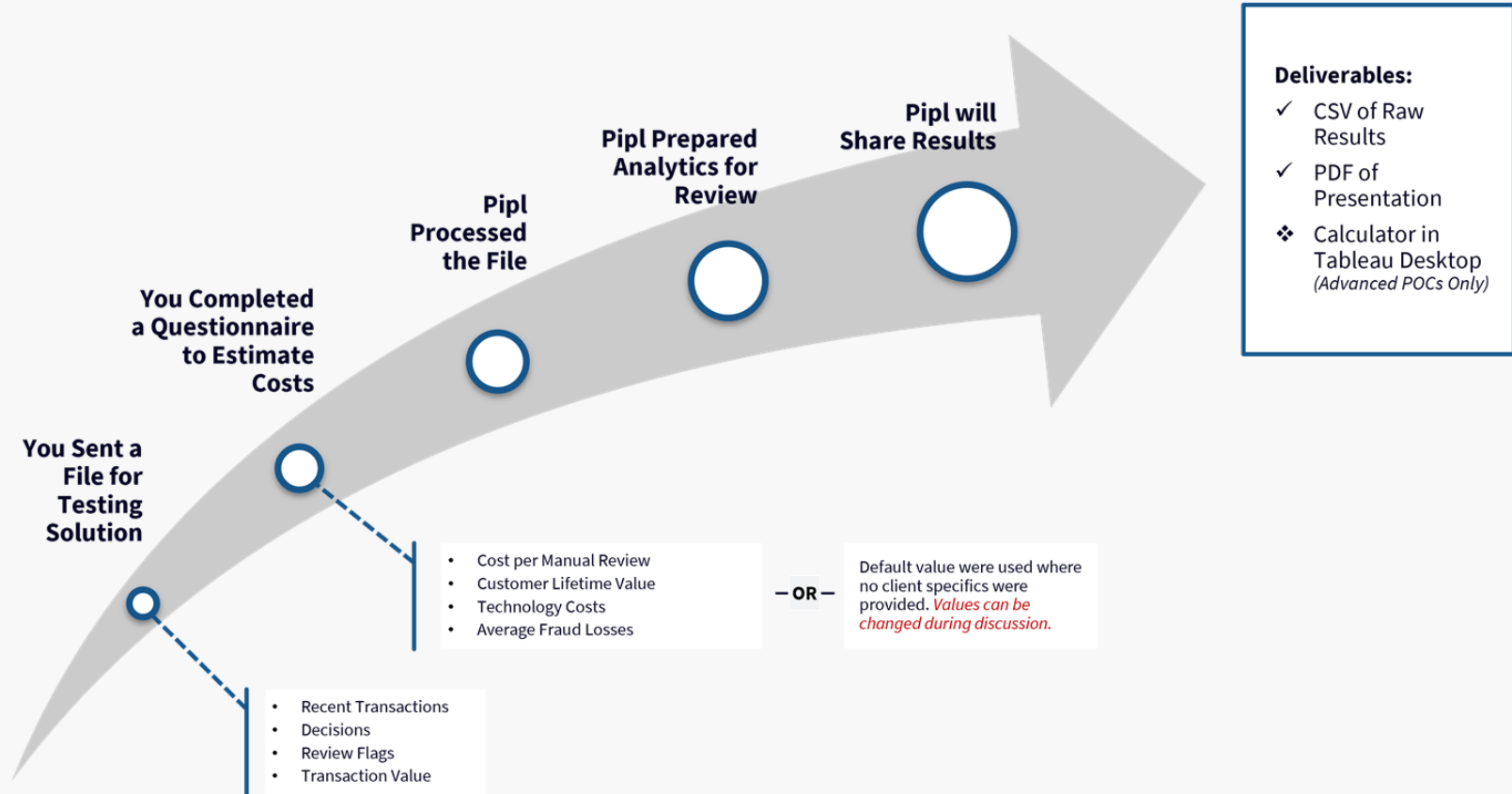


Pipl Trust API

Predictive trust scoring enabling automated decisioning across the customer lifecycle.

The POC Process & Deliverables

Walking through the Proof of Concept Process and Files to be Returned



Overview of Inputs

Transactions Provided by the Customer for a PipI Trust Proof of Concept

Metric	Email Addresses	Phone Numbers
Number of Inputs Provided Total Rows Provided	192,750	0
Number of Errors Rows with Missing Inputs for Verification	810	0
Number of Inputs Verified Inputs with a Trust Score Returned (Rows without Errors)	191,940	0
Duplicate Inputs Values Submitted in Multiple Rows	24,536	0
Distinct Inputs Analyzed Actual Values with Duplicates and Errors Removed	167,404	0

Summary of Client Decisions

An overview of the final Decisions made by the customer at transaction time or following manual review.

Pipl's Trusted Intelligence (TI) helps customers improve automated decisions and reduce operational costs.

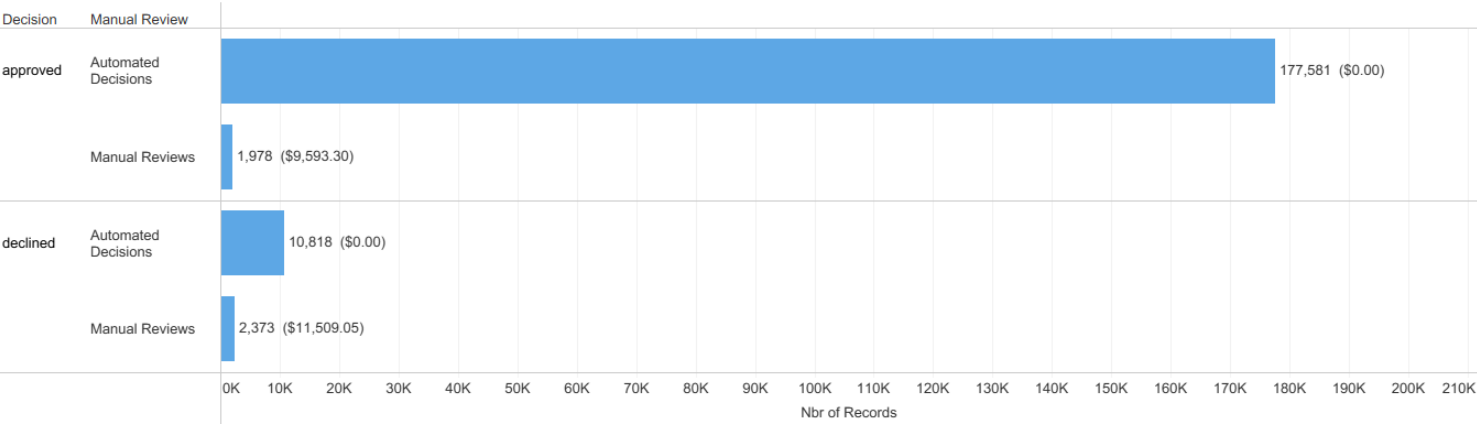
Estimated Spend on Manual Reviews

\$21,102.35

Estimated Cost per Manual Review

\$4.85

Summary of Customer Decisions
(with Estimated Costs Spent for Manual Review)



Decision	Automated Decisions						Manual Reviews					
	Distinct Email Addresses	Inputs Validated	Nbr of Records	% of Total	Estimated Review Costs	Estimated Technology Costs	Distinct Email Addresses	Inputs Validated	Nbr of Records	% of Total	Estimated Review Costs	Estimated Technology Costs
approved	156,430	176,844	177,581	92.13%	\$21,531.70	\$1,065,486.00	1,496	1,977	1,978	1.03%	\$239.83	\$11,868.00
declined	8,557	10,746	10,818	5.61%	\$1,311.68	\$64,908.00	1,270	2,373	2,373	1.23%	\$287.73	\$14,238.00
Grand Total	164,775	187,590	188,399	97.74%	\$22,843.38	\$1,130,394.00	2,761	4,350	4,351	2.26%	\$527.56	\$26,106.00



Overview of **Trust Signals**

Definitions for Returned Data



Attribute Terminology

Trust Signals Included in Returned Data

is_valid

Email Validity

The syntax of the email address is valid

Values: *true, false*

age_months

Email Age in Months

The number of months since the email was added to the Pipl database

Values: *Any integer*

domain_type

Domain Type

The email domain class type

Values: *disposable, commercial, personal, other*

number_of_sources

Number of Sources

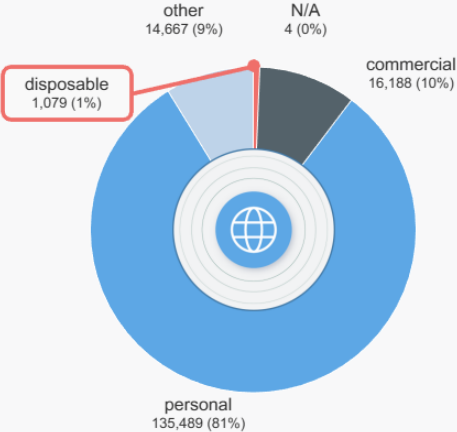
The number of sources that reported the email address

Values: *Any integer*

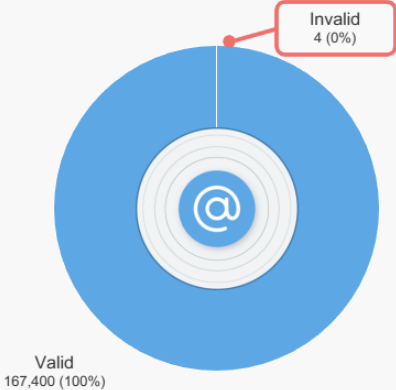
Pipl Trust Attribute Volumes

Where the POC results falls for each Trust Signal

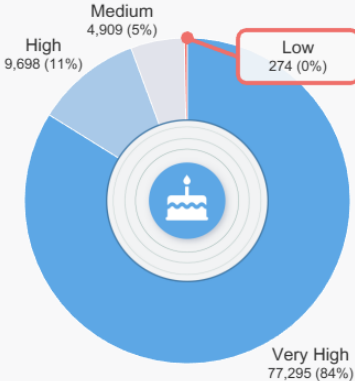
Domain Type Summary



Email Validity Summary



Email Age (Text)



Types disposable commercial personal other N/A

Validity Invalid Valid

Email Age Very High High Medium Low

1,079

email addresses processed
were profiles created through a
disposable domain

99.998%

email addresses provided
contained a valid format following
RFC and IETF standards

274

emails less than 6 months old.
All emails ranged between
3 months and 122 years

What Your Trust Score Means

Pipl Trust scores are numbers between 0-1000 representing trust of a specific item (Email, Phone) or a specific action (Purchase, Subscribe, Signup, etc.)

CATEGORY	SCORE RANGE	DESCRIPTION
Complete Trust	800 - 1000	• Email address belongs to a trusted identity • Email has been seen across multiple sources
Strong Trust	550 - 799	• Email address likely links back to a trusted identity • Contains some trust signals • Investigate further, depending on risk tolerance
Moderate Trust	300 - 549	• Identity trustworthiness is questionable • Not enough trust signals • Send to manual review
Low Trust	200 - 299	• Lack of trust signals indicates identity is not trustworthy • Email could be newly created and/or disposable with a risky domain type
No Trust	0 - 199	• Email is not trustworthy • Email is likely fraudulent

EmailTrust Score Distribution

Use the sliders at the bottom to adjust the threshold for automatically approving or rejecting customers based on their PipI Trust score attribute.

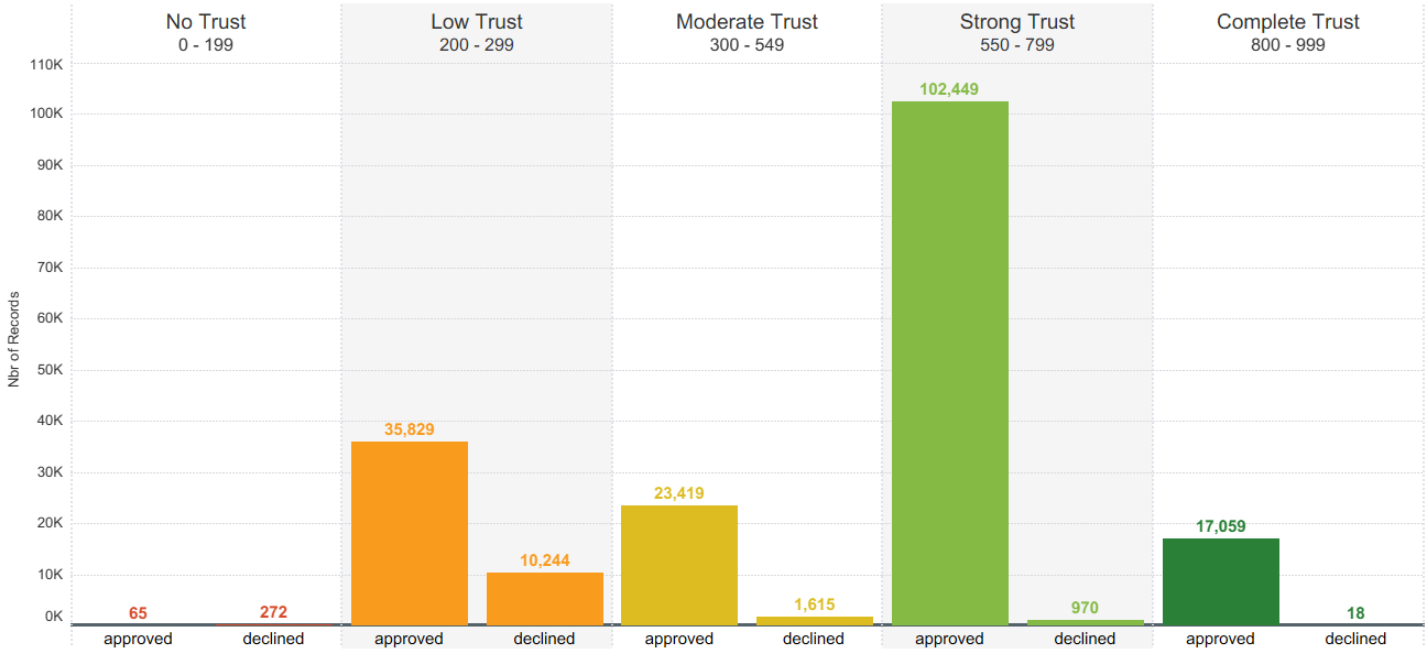
Our POC Evaluation tools help you navigate early operational workflow and make adjustments to maximize effectiveness.

Analyze our Scores and compare them to your historical decisions and fraud outcomes to determine thresholds that streamline your workflow.



Adjust Workflow to Incorporate Score in Decisioning

Calculate impact of thresholds to automatically approve or deny transactions and to minimize impact to review queue.



Automatically Deny Transactions **Beneath** a Desired Trust Score



Automatically Approve Transactions **Above** a Desired Trust Score





Exploring **Trends & Insights**

Notable Patterns in POC Data



Score Distribution by Domain Type

Each domain type provides insight into your customer base and can improve automated and human decisions.

Operational costs incurred from manually reviewing the activities of disposable profiles can be substantial.

Domain Type can aid in optimizing business rule decision engines and in tuning machine learning models. Explore total reviews for disposable domains.

Estimated Cost per Manual Review

\$4.85

% of Transactions that Require Manual Review

2.5%

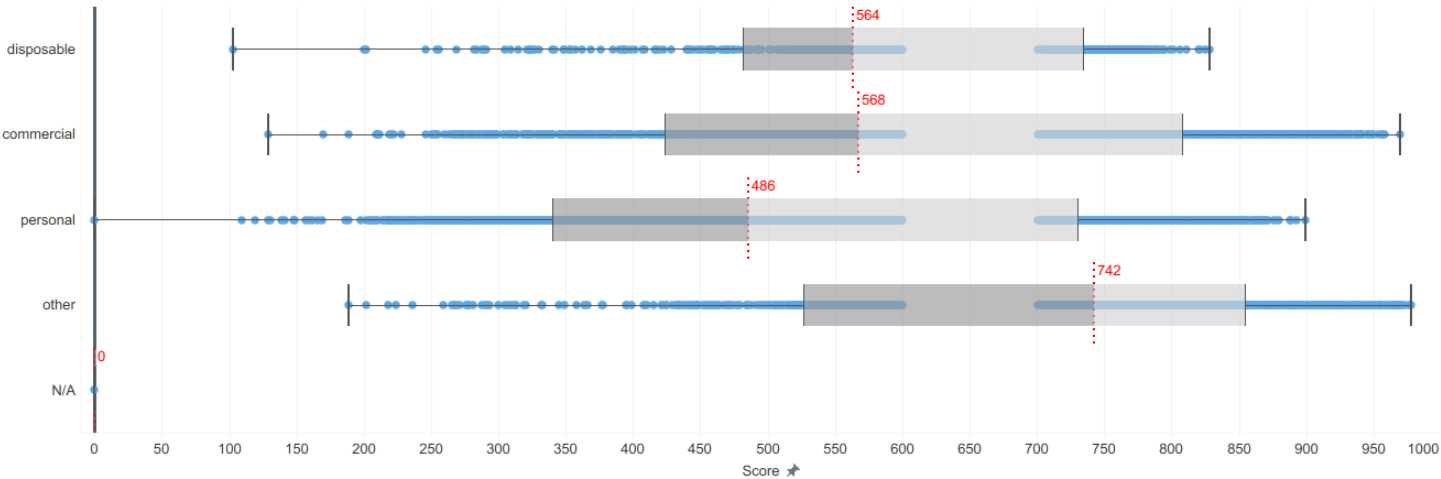
Estimated Technology Cost per Transaction

\$6.000

Estimated Monetary Loss Cost per Account

\$35.00

Domain Type Distribution with Median Trust Score



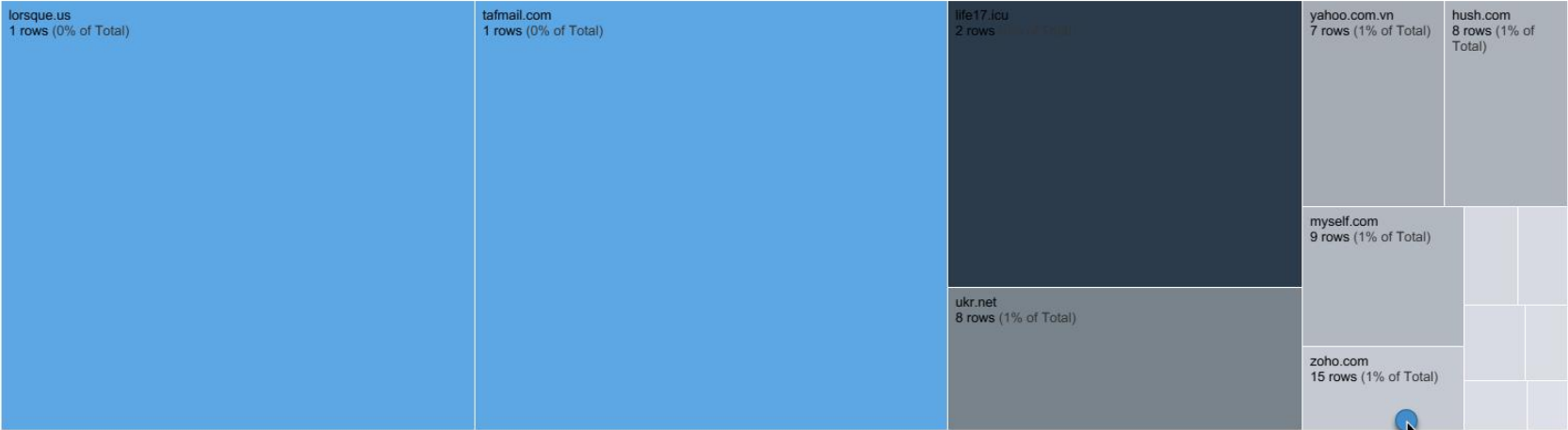
Domain Type	approved						declined					
	Nbr of Records	% of Total	Total Reviews	Estimated Review Costs	Estimated Technology Costs	Estimated Monetary Losses	Nbr of Records	% of Total	Total Reviews	Estimated Review Costs	Estimated Technology Costs	Estimated Monetary Losses
disposable	1,002	0.52%	12	\$121.49	\$6,012.00	\$35,070.00	256	0.13%	14	\$31.04	\$1,536.00	\$8,960.00
commercial	18,758	9.77%	275	\$2,274.41	\$112,548.00	\$656,530.00	645	0.34%	103	\$78.21	\$3,870.00	\$22,575.00
personal	141,951	73.96%	1,471	\$17,211.56	\$851,706.00	\$4,968,285.00	11,816	6.16%	2,230	\$1,432.69	\$70,896.00	\$413,560.00
other	17,108	8.91%	219	\$2,074.34	\$102,648.00	\$598,780.00	400	0.21%	26	\$48.50	\$2,400.00	\$14,000.00
N/A	2	0.00%	0	\$0.24	\$12.00	\$70.00	2	0.00%	0	\$0.24	\$12.00	\$70.00
Grand Total	178,821	93.17%	1,977	\$21,682.05	\$1,072,926.00	\$6,258,735.00	13,119	6.83%	2,373	\$1,590.68	\$78,714.00	\$459,165.00

Explore Trends in Domain Types

Explore Domain Types of interest. Click any type to see the list of email addresses submitted, their Trust Signals, and the client's final decision.

Domain Types (disposable) by Percent Reviewed

Click any domain to see the emails for this type in the table below.



List of All Email Addresses

with Average Score, Number of Records submitted, and Summary of Customer Decisions

Email Address	Email Domain	Is Valid	Age (Months)	Number Of Sources	Avg. Score	Nbr of Records	Total Approved	Total Declined	Total Reviews	Estimated Technology Cost per Transaction
	zoho.com	Valid	0	0	282	1	1	0	0	
	zoho.com	Valid	0	0	200	1	0	1	0	
	zoho.com	Valid	0	0	200	1	0	1	0	
	zoho.com	Valid	0	0	200	1	1	0	0	
	zoho.com	Valid	0	0	200	1	1	0	0	
	zoho.com	Valid	0	0	200	1	1	0	0	
	zoho.com	Valid	0	0	515	1	1	0	0	\$0.12

zoho.com
15 rows (1% of Total)
15 Distinct Email Addresses
Type: disposable

Customer Decisions
Approved: 12 (80%)
Declined: 3 (20%)
Reviewed: 1

Trust Scores
Average Score: **318.3**
Median Score: **200.0**

Interactive Controls:

Filter by Decision

- ☒ (All)
☐ approved
☐ declined

Choose a Domain Type:

- ☐ (All)
☒ disposable
☐ commercial
☐ personal
☐ other
☐ N/A
☐ Null

Choose a Measure to

Prioritize Domains:

- ☐ Distinct Email Addresses
☐ Average Score (low to h...
☐ Average Score (high to ...
☐ Median Score
☐ Percent Invalid
☐ Percent Valid
☐ Percent Approved
☐ Percent Declined
☒ Percent Reviewed

Estimated Technology Cost per Transaction

\$6.000

Estimated Monetary Loss per Account

\$35.00

Legends:

Selected Measure

0 1



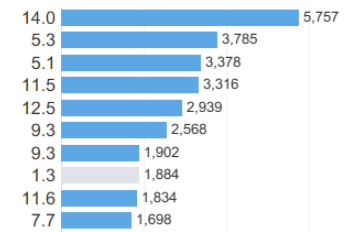
Email Age Trends

EmailTrust is the first line of defense to establish trust in a transaction

Email Addresses over 4 years old seen from a large number of sources have high trust.

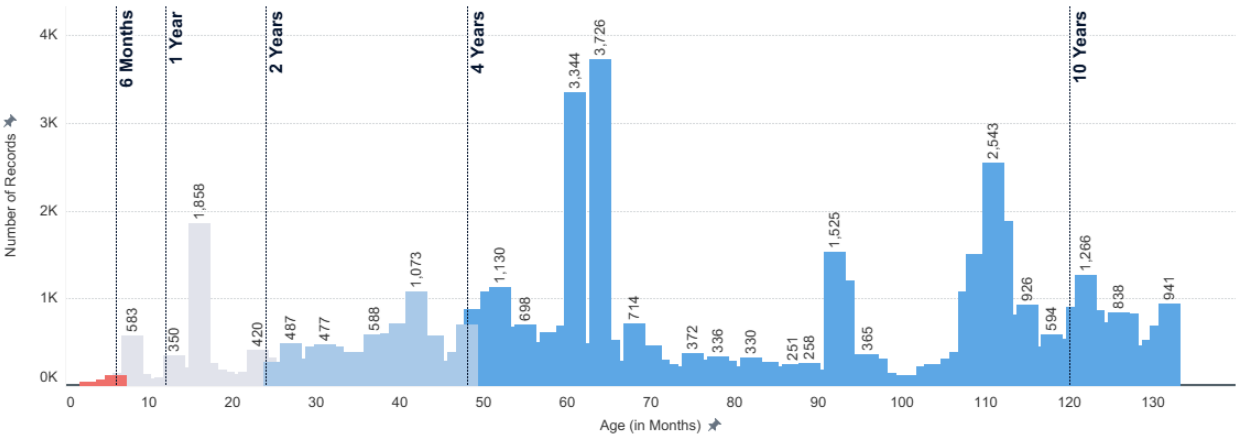
You can strengthen automated decisions, reduce fraud losses, and minimize cost of human review and manual research using PipI Trust insights.

Top 10 Most Frequently-Seen Email Ages
Value in Years



Email Age Distribution by Final Decision

Change the number of years in display to hone in on smaller ranges.



Interactive Controls:

Set the Max Age in Years to Display in top chart

Filter by Decision:

- ☒ (All)
☐ approved
☐ declined

Estimated Cost per Manual Review

Estimated Technology Cost per Transaction

Estimated Monetary Loss per Account

Legends:

Email Age (text)

- ☒ Very High
☒ High
☒ Medium
☒ Low

		approved						declined					
Email Age (text)	Email Age (range)	Nbr of Records	% of Total	Total Reviews	Estimated Review Costs	Estimated Technology Costs	Estimated Monetary Losses	Nbr of Records	% of Total	Total Reviews	Estimated Review Costs	Estimated Technology Costs	Estimated Monetary Losses
Very High	Over 4 years	86,075	44.66%	1,190	\$10,437	\$516,450	\$3,012,625	978	0.51%	136	\$119	\$5,868	\$34,230
High	2 - 4 years	10,872	5.64%	139	\$1,318	\$65,232	\$380,520	252	0.13%	47	\$31	\$1,512	\$8,820
Medium	7 months to 2 ..	5,367	2.78%	44	\$651	\$32,202	\$187,845	241	0.13%	22	\$29	\$1,446	\$8,435
Low	0 - 6 months	287	0.15%	4	\$35	\$1,722	\$10,045	10	0.01%	0	\$1	\$60	\$350
Age Unavail..	Uncategorized	76,958	39.93%	601	\$9,331	\$461,748	\$2,693,530	11,710	6.08%	2,168	\$1,420	\$70,260	\$409,850
Grand Total		179,559	93.16%	1,978	\$21,772	\$1,077,354	\$6,284,565	13,191	6.84%	2,373	\$1,599	\$79,146	\$461,685

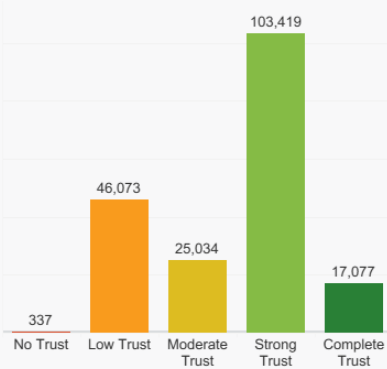
Adjusting for Trust Score Automated Decisions

Transactions Score Ranges with Optimization of Automated Decisions

Identify thresholds of high and low trust that can increase automated decisions with confidence.

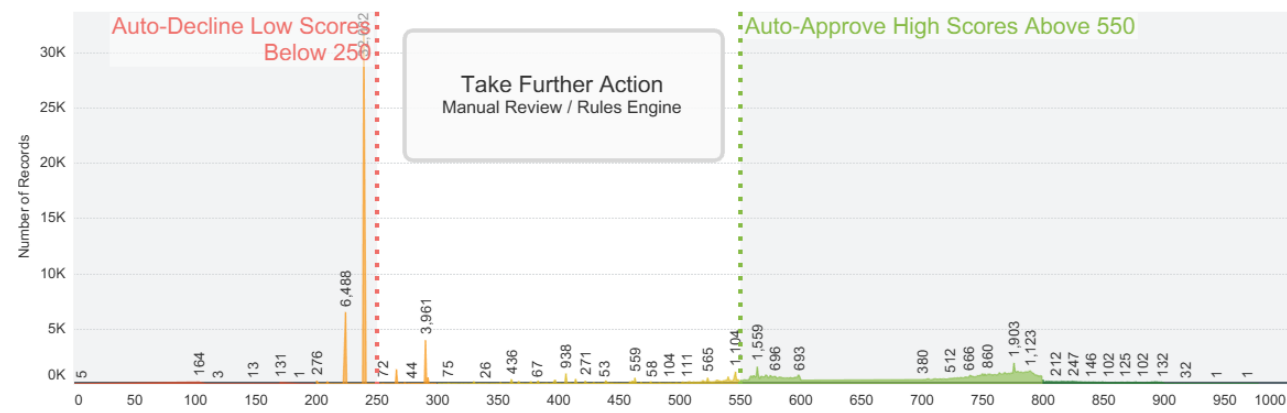
Scores in the middle range can continue downstream, lead to further review, or flow into a business rules engine for further analysis.

Summary of Transactions by Score Band



Adjust Workflow to Incorporate Score in Decisioning

Calculate impact of thresholds to automatically approve or deny transactions and to minimize impact to review queue.



Optimize Decisions in Manual Review

Automatically Deny **Beneath** this Score

250

Automatically Approve **Above** this Trust Score

550

Total Automated Decisions

Auto-Denials

39,477

1,996 Reviewed

Take Further Action (Review)

31,967

Auto-Approvals

120,496

1,622 Reviewed

Interactive Controls:

Decision

☒ (All)

☐ approved

☐ declined

Manual Review

☒ (All)

☐ Automated Decisions

☐ Manual Reviews

Estimated Technology Cost per Transaction

\$6.000

Estimated Cost per Manual Review

\$4.85

Estimated Monetary Loss per Account

\$35.00

Legends:

Score Band Category

■ No Trust

■ Low Trust

■ Moderate Trust

■ Strong Trust

■ Complete Trust



Performance Measures

Recommendations to Maximize Uplift



Enter Costs for Calculations

Use our interactive calculator to estimate costs associated with our Trust Indicators and Scores.

ROI Calculator Inputs:

Estimate your business costs to better calculate impact to the bottom line and find insights.

Total Records in POC

192,750

Number of Inputs with Trust Score

191,940

Distinct Email Addresses

167,404

Enter Customer Costs and Workflow

Enter the costs provided in the customer questionnaire or estimated values for uplift calculations.

Average Technology Cost per Transaction

What are your average operational cost incurred for each account or transaction that require a non-automated decision via manual review or additional research?

Default = \$6

With Trusted Intelligence at the top of your funnel of identity and fraud services called per transaction, estimate the total costs spent per transaction for those technologies. Exclude existing in-house decisioning solutions.

Average Technology Cost per Transaction

\$6.000

Average Cost per Manual Review

What is your average operational cost incurred per manual review (additional human decisioning)? This cost is dependent upon the number of reviewers, salaries, and throughput.

Default = 4% at \$4.85 per review

This value allows us to estimate the costs associated with any transactions that were flagged as Reviewed = True. Where no review indicators were provided, we can estimate the percentage of transactions that will require review instead. As we target score ranges for automated decisions, we can calculate cost reductions.

% of Transactions that Require Review

2.5%

Average Cost per Manual Review

\$4.85

Average Cost of False Declines

False declines are legitimate transactions that are incorrectly declined by the online business. These are also known as false positives. According to the Merchant Risk Council's *Global Fraud Survey*, the average online store declines 2.6% of all orders because of false declines.

Default = 3.1% at \$100 per account

Higher purchasing prices lead to higher percentages of declines. This value allows us to estimate the financial impact of false declines. Based on the lifetime value of your customers and this rate, we can estimate loss in revenue and loyalty that can be avoided by automatically approving complete trust scores.

% of False Declines

3.1%

Customer Lifetime Value

\$100.00

Average Fraud Loss per Account

What is your average loss per Account? For all accounts that are identified as fraudulent or created by synthetic identities, what is average financial loss associated with those accounts?

Default = 3.5% at \$35 per account

This value allows us to estimate the potential losses associated with customers in low trust bands. Based on your overall rate of accounts going bad and the average loss per account, we can estimate savings from automatically declining low-trust accounts up front.

Global Fraud Rate

3.5%

Average Monetary Loss per Account

\$35.00

ROI & Uplift Summary

Estimated Return on Investment Based on Automated Decisions

This calculator uses your chosen score ranges to estimate the quantitative and monetary impact of today's analysis.

Impact of Automated Decisions

Auto-Declines

39,477

1,996 Reviewed

Auto-Approvals

120,496

1,622 Reviewed

Total Uplift

\$1,017K

1	2	3	4	5	Quantity	Amount
	Reduction in Technology Costs Transactions selected for <u>auto-approve</u> or <u>auto-decline</u> that no longer need to call additional downstream services for identity verification or fraud prevention. <i>Calculation: Transactions with scores above 550 and scores below 250.</i>				159,973	\$959,838
	Reduction in Manual Reviews Transactions selected for <u>auto-decline</u> that you confirmed were manually reviewed. Low Trust Scores will prevent the need for research. <i>Calculation: Manual Review = True with scores below 250</i>				1,996	\$9,681
	Improved Customer Experience Transactions selected for <u>auto-approval</u> that you confirmed were manually reviewed. High Trust Scores will expedite the review process and reduce customer friction. <i>Calculation: Manual Review = True with scores above 550.</i>				1,622	\$7,867
	Reduction in False Declines Transactions selected for <u>auto-approval</u> but were declined originally due to suspected fraud but were falsely declined. <i>Calculation: Transactions with scores above 550 and Decision = declines multiplied by the percentage of false declines (3.1%).</i>				31	\$3,100
	Fraud Losses Avoided Up-Front Transactions selected for <u>auto-decline</u> whose low Trust Scores allowed you to confidently avoid known financial losses and potential fraud exposure. <i>Calculation: Transactions with scores below 250 that have a reported amount or were not already declined.</i>				1,033	\$36,156

Interactive Controls:

Automatically Deny Beneath this Score

Automatically Approve Above this Score

Estimated Technology Cost per Transaction

% of Transactions that Require Manual Review

Estimated Cost per Manual Review

% of False Declines

Customer Lifetime Value

Global Fraud Rate

Estimated Monetary Loss per Account

ROI & Uplift Summary

Estimated Return on Investment Based on Automated Decisions



\$1,016,641

Total Uplift

Monetary uplift in reduced technology costs, reduced operational costs of human review, and fraud losses avoided through improved automated trust decisions.

164,655

Automated Decisions



\$96,375

Total Spend

Total cost of service based upon the number of transactions processed in Proof of Concept and billed amount per API call / transaction (0.50).

192,750

POC Transactions



\$920,266

Bottom Line Savings

Savings estimated from using Pipl's Trusted Intelligence to confidently automate decisions on digital identity trust in your workflow. Estimates based on the spend per transaction in a Proof of Concept

11 : 1

ROI Ratio



© 2022 Pipl. All rights reserved. Pipl and the Pipl logo are trademarks or registered trademarks of Pipl.

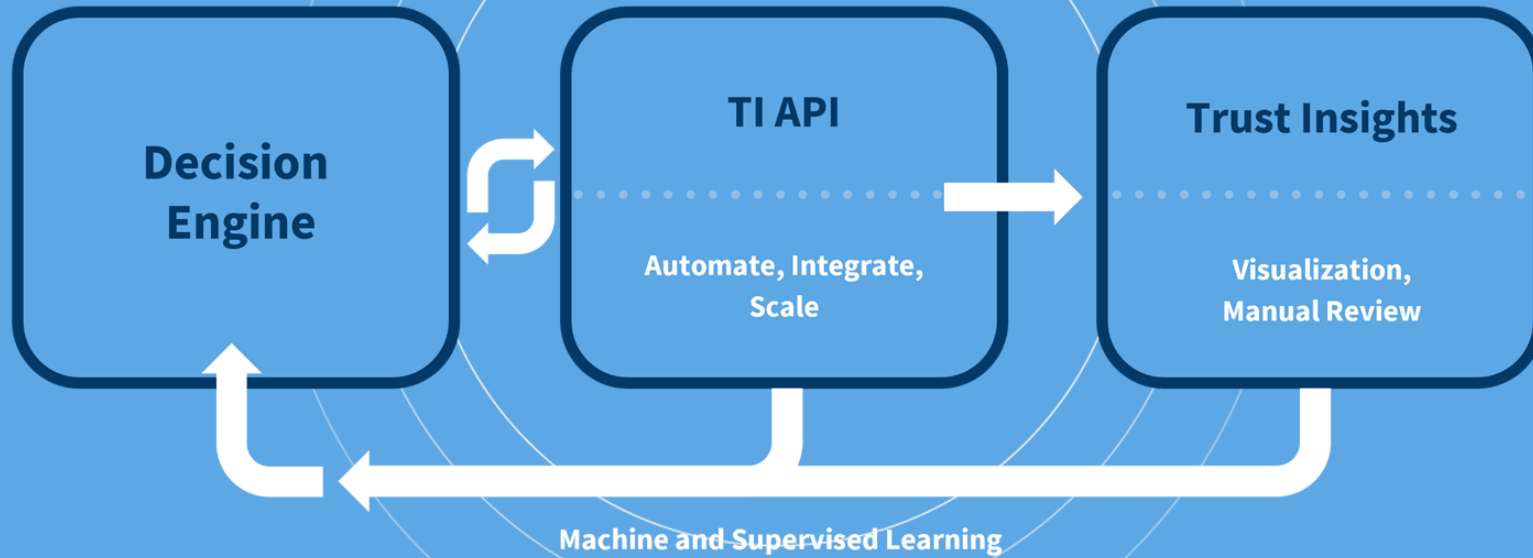


Solutions & Roadmap

Next Steps



End-to-end API Flow



The power of trusted identities **in your application**

- Automation
- Scale
- Only the data you need
- Instant access
- Seamless integration with your systems





Thank You

pipl.com

